

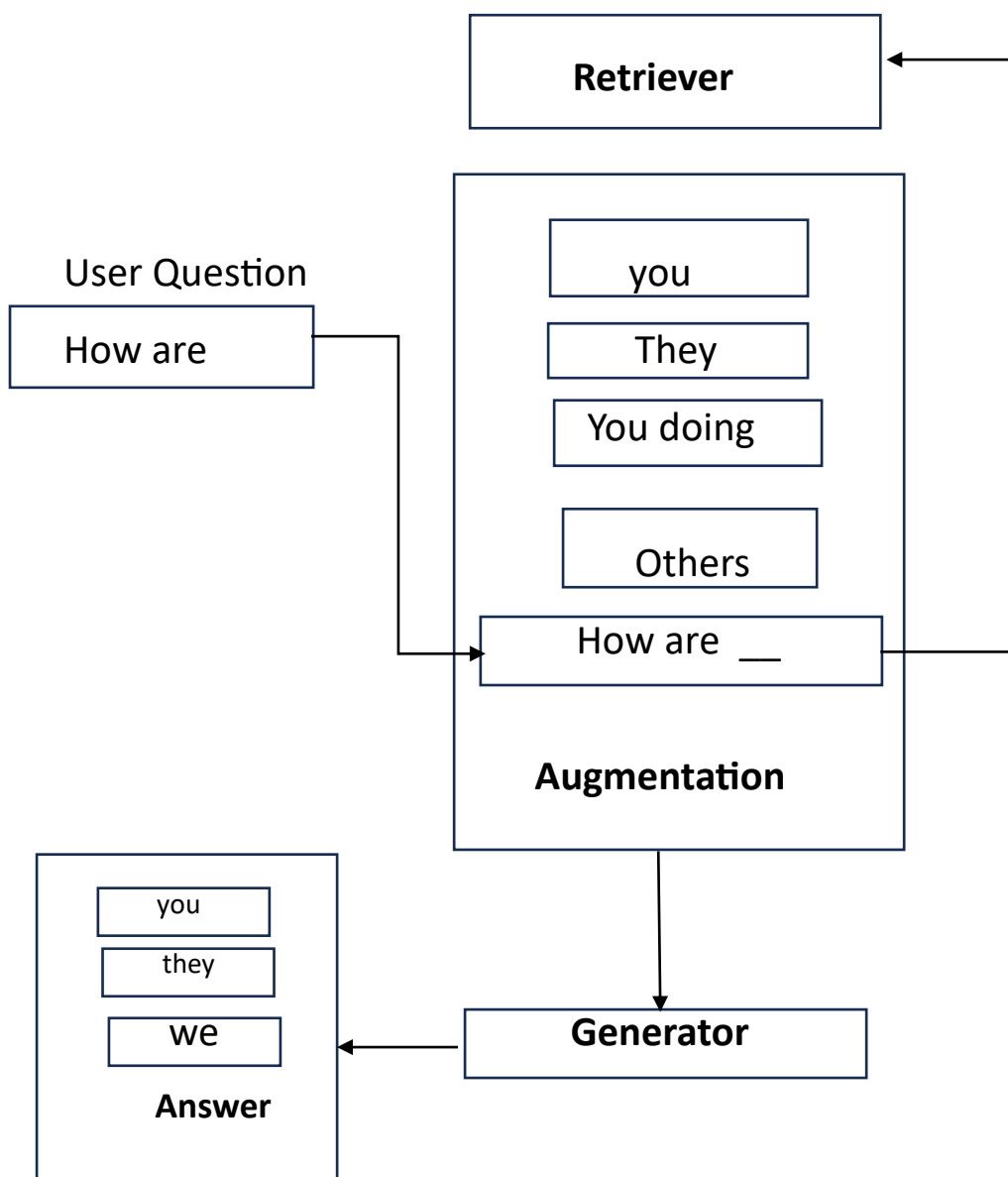
LangFuse with LLM for RAG

Langfuse is an open-source LLM engineering platform designed to streamline quality management by integrating Language Model-based feedback mechanisms. By leveraging LangFuse, organizations can ensure continuous quality improvement and mitigate potential side effects of software changes through real-time user feedback analysis.

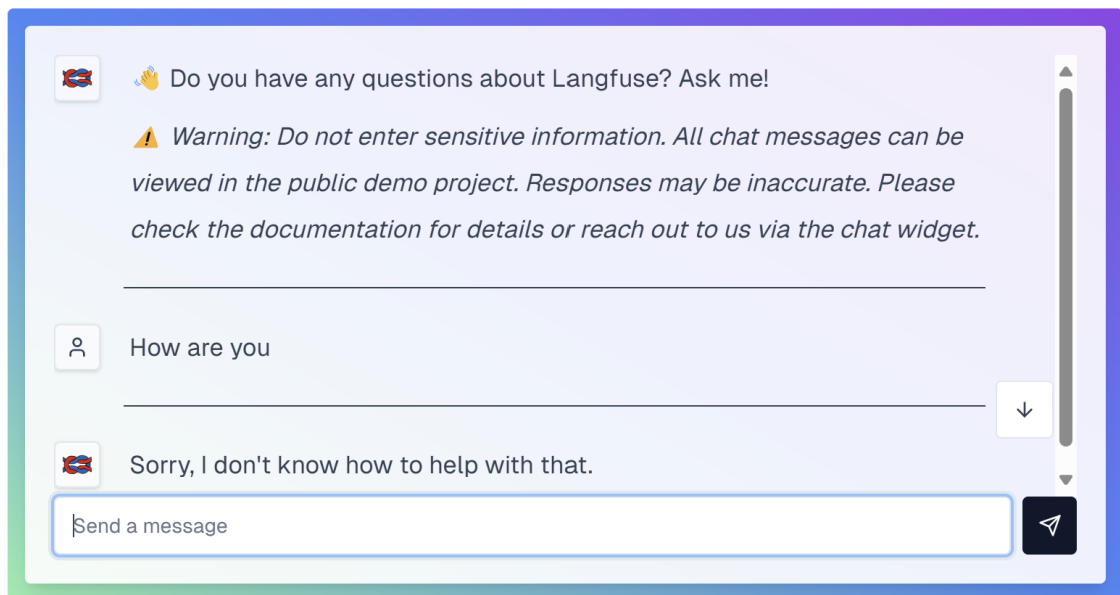
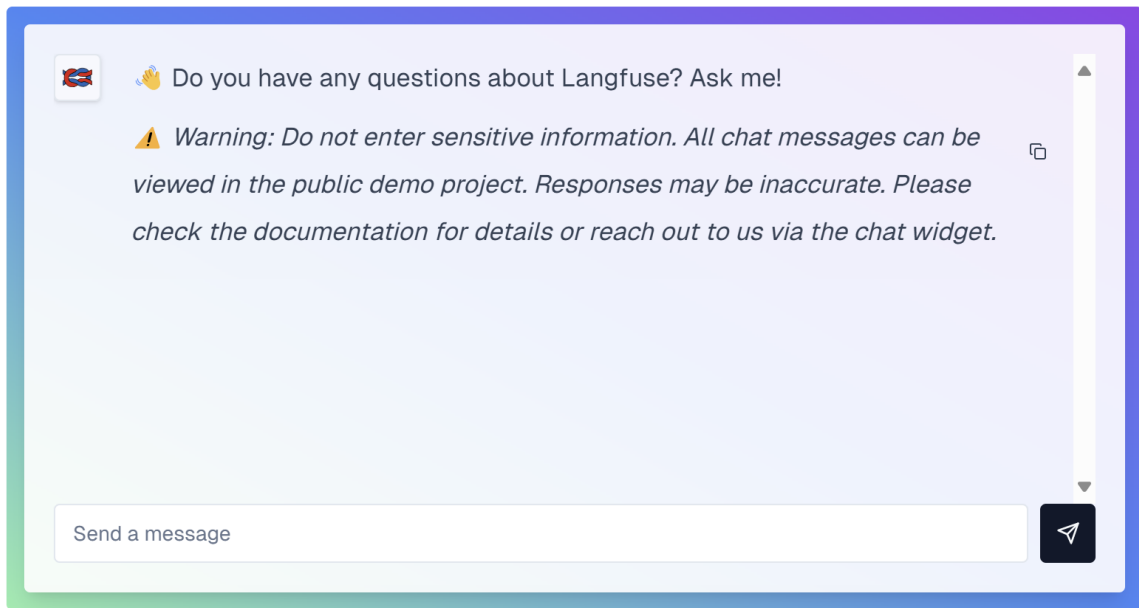
Langfuse has been especially useful in three areas.

- 1) helping us improve LLM output.
- 2) debugging issues and errors .
- 3) providing more clarity on cost, usage and latency.

Basic RAG Workflow



Langfuse – Bot



Langfuse - TRACES

LLM applications use increasingly complex abstractions, such as chains, agents with tools, and advanced prompts. **Nested traces** in Langfuse help to understand what is happening and identify the root cause of problems.

Traces

Search by id, name, user id		Q	Filter	1 hour	(17/29)			
☐	ID	Timestamp ▾	Name	User	Session	Latency	Usage	# correctness-v1
☐	☐ 022ab14b...	29/08/2024, 03:19:07	qa	☐ u-Gmzpkcb	☐ lf.docs.conversation....	2.74s	126 → 112 (Σ 238)	0.2000 ☐
☐	☐ feaeed11...	29/08/2024, 03:18:53	qa	☐ u-Gmzpkcb	☐ lf.docs.conversation....	1.33s	97 → 11 (Σ 108)	0.8000 ☐
☐	☐ fd284839...	29/08/2024, 03:13:11	qa	☐ u-kRNZKvo	☐ lf.docs.conversation....	2.51s	94 → 11 (Σ 105)	1.0000 ☐
☐	☐ 0a6a5ded...	29/08/2024, 02:37:19	qa	☐ u-VGqN2xu	☐ lf.docs.conversation....	2.31s	232 → 37 (Σ 269)	0 0.1000
☐	☐ 68851f70...	29/08/2024, 02:37:04	qa	☐ u-VGqN2xu	☐ lf.docs.conversation....	2.14s	181 → 36 (Σ 217)	0.1000 ☐
☐	☐ 63db8a95...	29/08/2024, 02:37:04	qa	☐ u-VGqN2xu	☐ lf.docs.conversation....	1.00s	185 → 0 (Σ 185)	0.1000 ☐
☐	☐ 81189607...	29/08/2024, 02:36:55	qa	☐ u-VGqN2xu	☐ lf.docs.conversation....	4.06s	140 → 34 (Σ 174)	0.2000 ☐
☐	☐ 2eb7e199...	29/08/2024, 02:36:36	qa	☐ u-VGqN2xu	☐ lf.docs.conversation....	2.60s	91 → 40 (Σ 131)	0.2000 ☐

Each Trace can contain multiple Observations to record individual steps of an execution. There are different types of Observations:

- Events are the basic building block. They are used to track discrete events in a Trace.
- Spans track time periods and include an end_time.
- Generations are a specific type of Spans which are used to record generations of an AI model. They contain additional metadata about the model, LLM token and cost tracking, and the prompt/completions are specifically rendered in the langfuse UI.

Trace Detail

☆ Private ⏮ ⏭ ⏹

Session: lf.docs.conversation.WmoUjUD User ID: u-kRNZKvo 94 → 11 (Σ 105) Total cost: \$0.00

Tags no-context

🌳 Tree 🕒 Timeline

TRACE qa

29/08/2024, 03:13:11

2.51s 94 → 11 (Σ 105) Release: 9cf087e7a5f100dcd213b0a474ee740258a6f010

🗑 📌 Annotate ➕ Add to dataset

Input

"how are"

Output

"Sorry, I don't know how to help with that."

Metadata

{
 pathname: "/docs/demo"

Preview

Scores

contextrelevance-v1: 0.00

correctness-v1: 1.00

hallucination-v2: 0.00

helpfulness-1: 0.00

language-detector-v3: 0.00

thank-you-v1: 0.00

toxicity-v2: 0.00

SPAN retrieval

1.32s

WARNING

GENERATION prompt-embedding

0.65s 2 → 0 (Σ 2)

DEBUG

SPAN vector-store

0.67s

SPAN context-encoding

0.00s

SPAN fetch-prompt-from-langfuse

Evaluation Metrics

In Langfuse, scores are objects that store evaluation metrics and are associated with traces. They can be attached to specific observations within a trace and can be used in multiple ways across the platform

Scores

EVAL

conciseness-v1: 1.00

contextrelevance-v1: 0.00

correctness-v1: 1.00

hallucination-v2: 0.00

helpfulness-1: 0.00

language-detector-v3: 0.00

thank-you-v1: 0.00

toxicity-v2: 0.00

Langfuse Dashboard

In langfuse you get a one stop view of LLM performance ,analyze and monitor the model usage and cost.

Dashboard

Aug 28, 2024

24 hours

Request Chart

Traces

154 Total traces tracked

qa154

Model costs

\$0.0376 Total cost

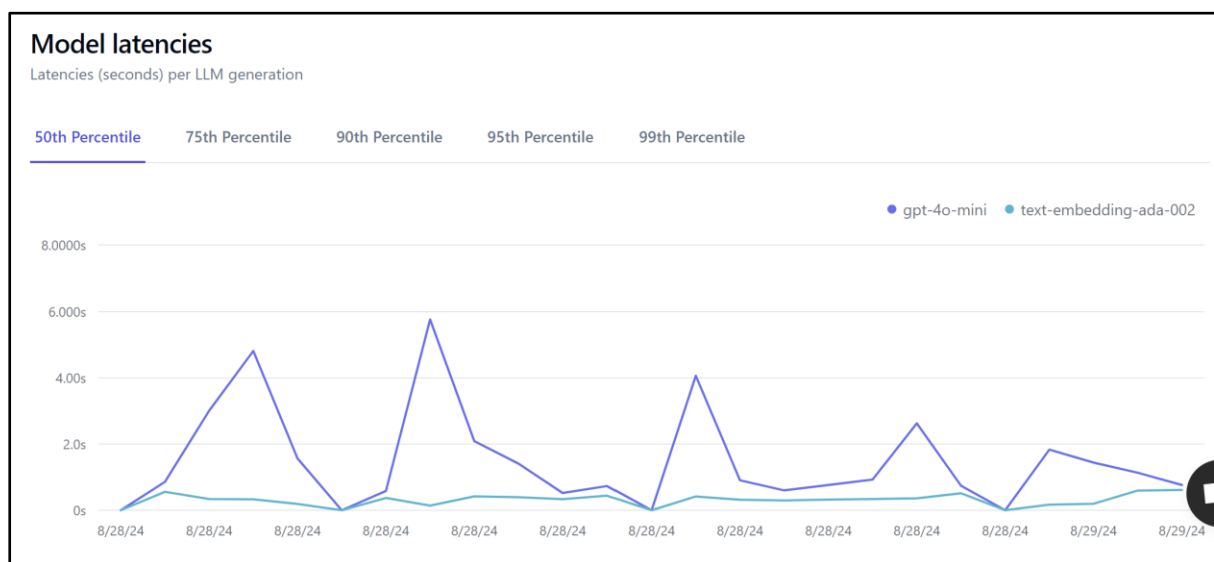
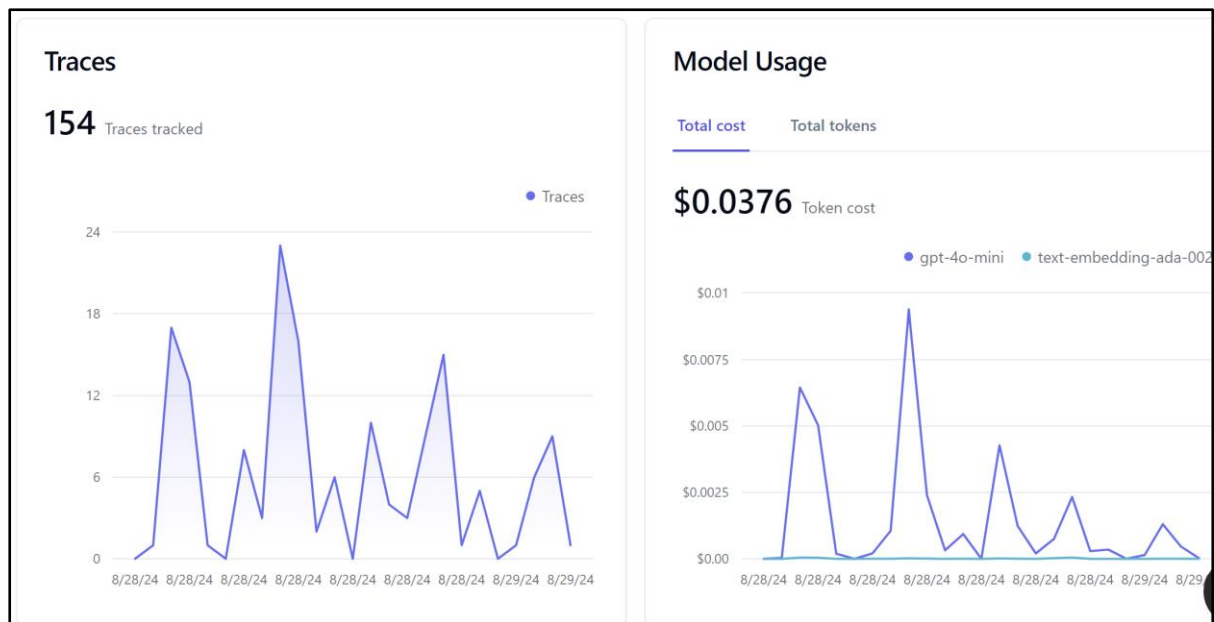
Model	Tokens	USD
gpt-4o-mini	181.54K	\$0.0374
text-embedding-ada-002	2.47K	\$0.0002

Scores

1.24K Total scores tracked

Name	#	Avg	0	1
# correctness-v1 (eval)	155	0.55	8	18
# language-detector-v3 (eval)	155	0.74	40	115
# contextrelevance-v1 (eval)	155	0.54	67	71
# toxicity-v2 (eval)	155	0	152	0
# conciseness-v1 (eval)	154	0.55	6	48

Show all



Below is google colab link of Python SDK used to send LLM data to Langfuse in a convenient way. It uses a worker Thread and an internal queue to manage requests to the Langfuse backend asynchronously. Hence, the SDK adds only minimal latency to your application. [python_sdk_low_level.ipynb - Colab \(google.com\)](#)

Use Case

Mava is an AI-first customer support platform built for community-first organisations. Hundreds of SaaS, Gaming, Web3 and open-source projects already use Mava to help support their communities.

Mava has an AI bot that can auto-respond to commonly asked questions in public community channels, such as Discord as well as a full omni-channel support dashboard for your whole team.

References

- <https://langfuse.com/guides/videos/introducing-langfuse-2.0>
- <https://langfuse.com/docs/demo>
- [Q&A Chatbot for Langfuse Docs - Langfuse Blog](#)
- <https://langfuse.com/faq/all/mava-langfuse-case-study>